

■ DirectedInfinity

`DirectedInfinity[ ]` represents an infinite numerical quantity whose direction in the complex plane is unknown.

`DirectedInfinity[z]` represents an infinite numerical quantity that is a positive real multiple of the complex number z .

You can think of `DirectedInfinity[z]` as representing a point in the complex plane reached by starting at the origin and going an infinite distance in the direction of the point z . ■ The following conversions are made:

`Infinity` `DirectedInfinity[1]`

`-Infinity` `DirectedInfinity[-1]`

`ComplexInfinity` `DirectedInfinity[ ]`

■ Certain arithmetic operations are performed on `DirectedInfinity` quantities. ■ In `OutputForm`,

`DirectedInfinity[z]` is printed in terms of `Infinity`, and `DirectedInfinity[ ]` is printed as `ComplexInfinity`.

■ See page 338. ■ See also: `Indeterminate`.